

Lesson 6: Machine Learning

45 minutes

Overview

In this lesson, students are introduced to the concepts of Artificial Intelligence and Machine Learning using the AI for Oceans widget. First students classify objects as either "fish" or "not fish" to attempt to remove trash from the ocean. Then, students will need to expand their training data set to include other sea creatures that belong in the water. In the second part of the activity, students will choose their own labels to apply to images of randomly generated fish. This training data is used for a machine learning model that should then be able to label new images on its own.

Standards

Full Course Alignment

CSP Conceptual Framework

- **DAT-2** - Programs can be used to process data, which allows users to discover information and create new knowledge.
- **IOC-1** - While computing innovations are typically designed to achieve a specific purpose, they may have unintended consequences.

CSTA K-12 Computer Science Standards (2017)

- **AP** - Algorithms & Programming

Agenda

Warm Up (5 minutes)

Activity (35 minutes)

Wrap Up (5 minutes)

Objectives

Students will be able to:

- Explain ways that designers and developers can consider the potential effects of their programs
- Reason about how human bias plays a role in machine learning.

Preparation

- Review and complete the online tutorial yourself. If you are not going to use AI for Oceans, explore the other options listed below.
- Check the **"Teacher's Lounge"** forum for verified teachers to find additional strategies or resources shared by fellow teachers
- If you are teaching virtually, consider checking our **Virtual Lesson Modifications**

Links

Heads Up! Please make a copy of any documents you plan to share with students.

For the teachers

- **CSP Unit 5 - Lesson 6 - Machine Learning** - Slides [▼ Make a Copy](#)

Vocabulary

- **Data Bias** - data that does not accurately reflect the full population or phenomenon being studied
- **Information** - the collection of facts and patterns extracted from data

Teaching Guide

Warm Up (5 minutes)

 **Discuss:** *How can machines "learn"?*

Have students brainstorm silently on their own, then have them share with neighbors, and finally have them share out with the room.

Discussion Goal: Based on yesterdays conversations, answers may vary. Steer the discussion towards conversations around the role that humans play in machines learning. It's ok if the discussion here is short - you are setting the stage for the upcoming activity.

Remarks

Today we're going to be learning more about Machine Learning and its impacts.

Activity (35 minutes)

 **Video:** Play the video "What is Machine Learning".

Teaching Tip

Alternatives to AI For Oceans: AI for Oceans was originally developed as an Hour of Code activity that can be completed by students with any device available. We have modified it for its usage here. Depending on your classroom situation, you might opt to replace the activity with:

- **Teachable Machine** - *Teachable Machine is a web-based tool that makes creating machine learning models fast, easy, and accessible to everyone. Teachable Machine is flexible – use files or capture examples live. It's respectful of the way you work. You can even choose to use it entirely on-device, without any webcam or microphone data leaving your computer.*
 - If your classrooms devices have cameras, Teachable Machines offers an engaging way to create training sets. Encourage students to teach the machine to represent rock, paper, or scissors with hand gestures. What are some possible ways for bias to enter in?
- **Machine Learning for Kids** - *This free tool introduces machine learning by providing hands-on experiences for training machine learning systems and building things with them. It provides an easy-to-use guided environment for training machine learning models to recognise text, numbers, images, or sounds.*
 - Machine Learning for Kids is a great option if your students want to work with text samples. Teach the machine to recognize words or passages that are happy or sad. Lots to play around with here!

Remarks

Machine learning refers to a computer that can recognize patterns and make decisions without being explicitly programmed. In this activity you're going to supply the data to train your own machine learning model. Imagine an ocean that contains creatures like fish, but also contains trash dumped by humans. What if we could train a computer to tell the difference and then use that technology to help clean the ocean?

 **Do This:** Direct students to Levels 2-4 on Code Studio. Students should spend around five minutes total on these levels. Prompt their thinking with the "Consider" on the slide. To program A.I., use the buttons to label an image as either "fish" or "not fish". Each image and label becomes part of the data used to train A.I. to do it on its

own. Once trained, A.I. will attempt to label 100 new images on its own, then present a selection that it determined have the highest probability of being "fish" based on its training. Students who consistently label things correctly should see an ocean full of different types of sea creatures, without much (or any) other objects.

Content Corner

Every image in this part of the tutorial is fed into a neural network that has been pre-trained on a huge set of data called **ImageNet**. The database contains over 14 million hand-annotated images. ImageNet contains more than 20,000 categories with a typical category, such as "balloon" or "strawberry", consisting of several hundred images. When A.I. is scanning new images and making its own predictions in the tutorial, it is actually comparing the possible categories for the new image with the patterns it found in the training dataset.

 **Discuss:** *How well did A.I. do? How do you think it decided what to include in the ocean?*

 **Video:** Play the video "Training Data & Bias".

 **Discuss:** *How do you think your training data influenced the results that A.I. produced?*

Discussion Goal: Get students to reflect on their experience so far. It is important at this point that they realize the labeling they are doing is actually programming the computer. The examples they show A.I. are the "training data".

Remarks

In the second half of the activity, you will teach A.I. about a word of your choosing by showing it examples of that type of fish. As before, A.I. doesn't start with any training data about these labels. Even though the words in this level are fairly objective, it's possible that you will end up with different results based on their training data. You might even intentionally train A.I. incorrectly to see what happens!

Content Corner

The fish in this tutorial are randomly generated based on some pre-defined components, including mouths, tails, eyes, scales, and fins, with a randomly chosen body color, shape, and size. Rather than looking at the actual image data, A.I. is now looking for patterns in these components based on how the student classifies each fish. It will be more likely to label a fish the same way the student would have if it has matching traits.

 **Do This:** Direct students to Levels 6-7 on Code Studio. Students should spend around five minutes total on these levels. Prompt their thinking with the "Consider" on the slide. Here, as before, students will use training data to teach A.I. to recognize different types of fish. The words in this list are intentionally more subjective than what students will have seen so far. Encourage students to decide for themselves what makes a fish look "angry" or "fun". Two students may choose the same label and get a very different set of results based on which fish traits were their focus. Encourage students to discuss their findings with each other or go back and choose new words. Each student will rely on their own opinions to train A.I. which means that A.I. will learn with the same biases held by the students. As students begin to see the role their opinion is playing, ask them to reflect on whether this is good or bad, and how it might be addressed.

 **Discuss:** *How could biased data result in problems for artificial intelligence? What are ways to address this?*

Discussion Goal: At this point, students should have some preliminary thoughts on how biased data leads to problems for artificial intelligence. They may bring up that if the data sets are trained incorrectly, there will be incorrect or misinterpreted conclusions. It can be addressed through diverse training sets. The following video dives into this subject further.

 **Video:** Play the video "How I'm fighting bias in algorithms" with Joy Buolamwini.

 **Discussions:**

- How can computing innovations which make use of Machine Learning reflect existing human bias?

- How could it be used to discriminate against groups of individuals?
- How can that bias be minimized?

Remarks

As we've seen, problems of bias are often created by the type or source of data being collected. Collecting more data does not mean that the bias is removed. Computing innovations can reflect existing human biases because of biases written into the algorithms or biases in the data used by the innovation.

Machine learning and data mining have led to innovations in medicine, business, and science but information discovered in this way has been used to discriminate against groups of individuals.

Programmers (that includes you!) should take action to reduce bias in algorithms used for computing innovations as a way to combat existing human biases. Be on the lookout! Bias can occur at any level in software development.

 **Review:** Play the video "Impact on Society" which recaps the concepts discussed today.

 1-7

AI for Oceans - Machine Learning



Wrap Up (5 minutes)

 **Discuss:** Which steps of this process do you think have to be done by humans? Would you be concerned if any of them were automated?

Time may be running short at this point in the class. Encourage students to share with a neighbor or share out with the room. The conversation should focus around bias.

 Teaching Tip

You can share these stories with your class to help them see how AI will impact the future.

- **Food Waste Is a Serious Problem. AI Is Trying to Solve It**
- **AI tech can identify genetic disorders from a person's face**
- **How an AI Startup Designed a Drug Candidate in Just 46 Days**
- **MIT AI tool can predict breast cancer up to 5 years early**
- **The Army steps up its pace on self-driving cars**
- **San Francisco says it will use AI to reduce bias when charging people with crimes**
- **AI is helping scholars restore ancient Greek texts on stone tablets**

Remarks

At this point, you've fully explored the core parts of the Data Analysis Process. Ultimately you are able to use the new information gained through visualizing and finding patterns (whether yourself or using Machine Learning) to make decisions. This is why being careful about bias is so important!

Assessment: Check For Understanding

Check For Understanding Question(s) and solutions can be found in each lesson on Code Studio. These questions can be used for an exit ticket.

Question: Think about examples of Machine Learning you may have encountered in the past such as a website that recommends what video you may be interested in watching next. Are the recommendations ever wrong or unfair? Give an example and explain how this could be addressed.



Check for Understanding



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